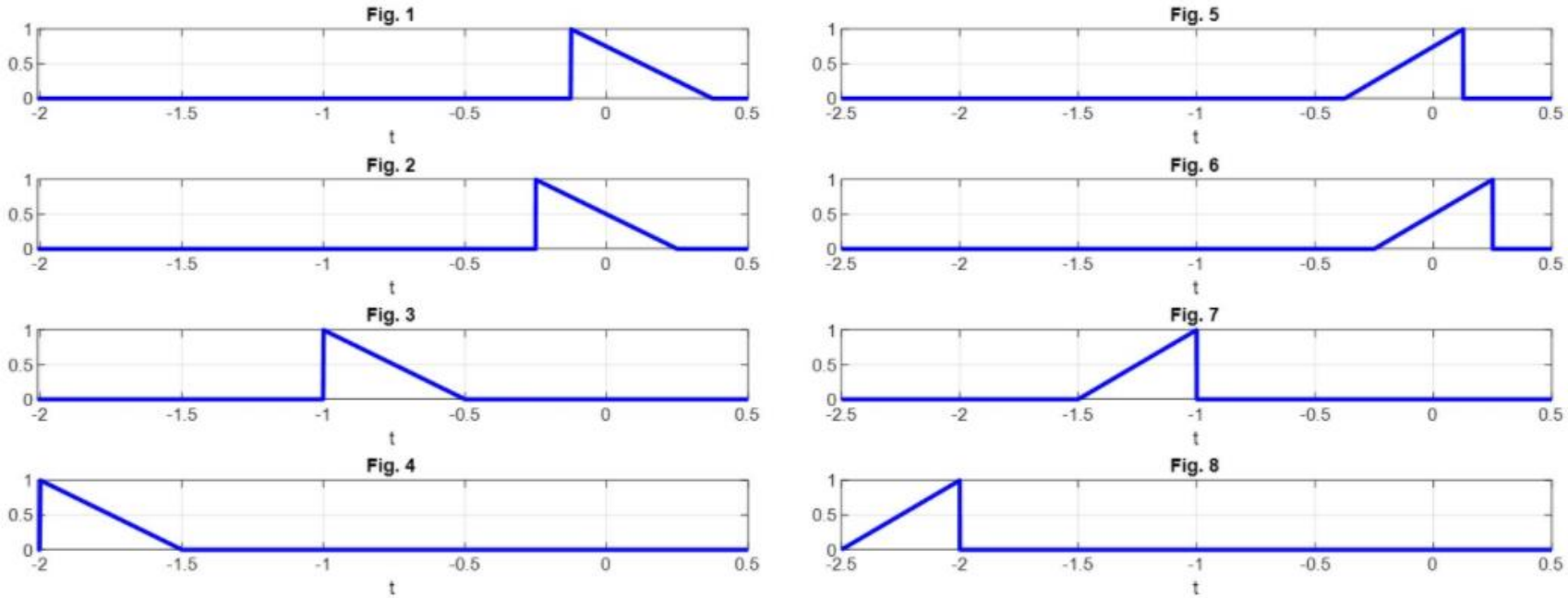
QUESTION 1: MULTIPLE CHOICE

10 out of 10 points

Consider the signal $x(t) = \mathrm{tri}\left(\frac{t}{4}\right) \times \mathrm{u}(t)$.

Which one of the below figures corresponds to the signal $x(8t+2)$?



Given Answer: Fig. 2

Correct Answer: Fig. 2

QUESTION 2: TRUE/FALSE

5 out of 5 points

The system $y(t) = x(t^2)$ is time invariant.

Given Answer: False

Correct Answer: False

QUESTION 3: TRUE/FALSE

5 out of 5 points

The system $y(t) = x\left(\frac{1}{t} - 3\right)$ is linear.

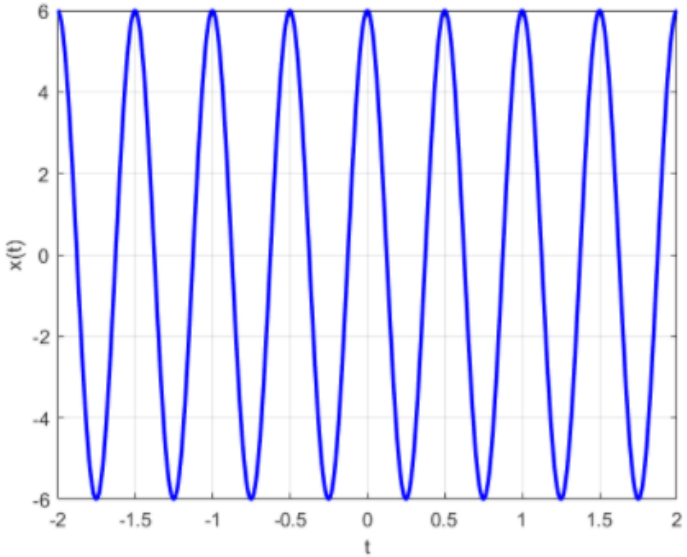
Given Answer: True

Correct Answer: True

QUESTION 4: CALCULATED NUMERIC

5 out of 5 points

What is the average power of the sinusoidal signal shown below?



Given Answer: 18

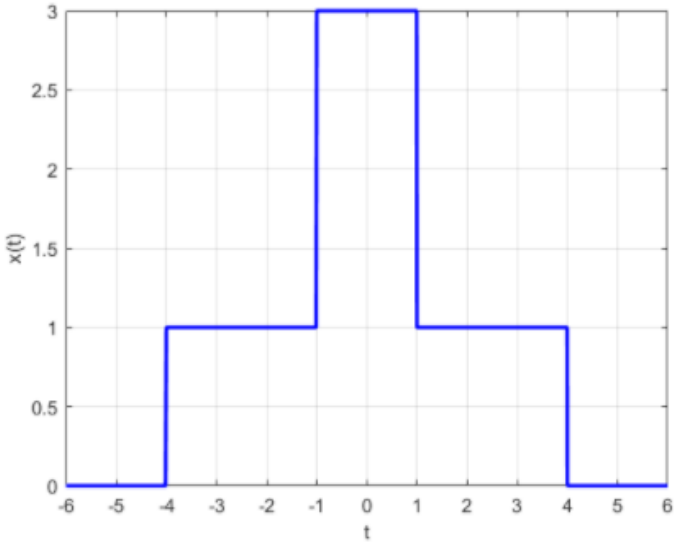
Correct Answer: 18

Answer range +/- 0 (18 - 18)

QUESTION 5: MULTIPLE CHOICE

10 out of 10 points

What is the energy of the signal shown in the figure below (in Joules)?



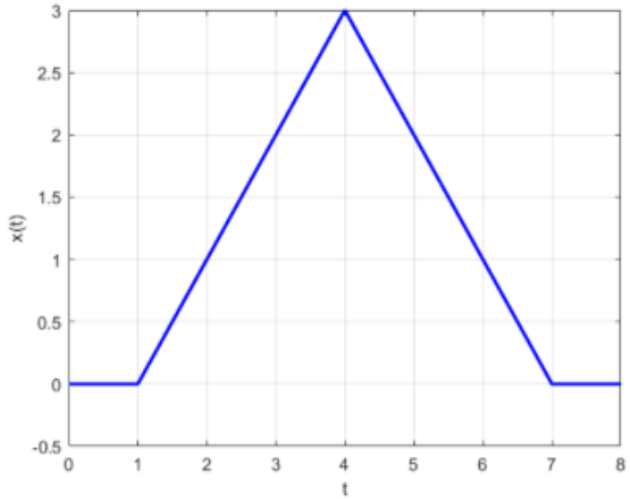
Given Answer: 24

Correct Answer: 24

QUESTION 6: MULTIPLE CHOICE

5 out of 5 points

If the signal below can be written as: a×ramp(t-b)-c×ramp(t-d)+e×ramp(t-f), where a, b, c, d, e and f are positive constants, what is the value of a?



Given Answer: 1

Correct Answer: 1

QUESTION 7: CALCULATED FORMULA

10 out of 10 points

Consider the signal: $x(t) = \text{rect}\left(\frac{t}{4a}\right) + 4\text{rect}\left(\frac{t-2a}{4a}\right)$.

What is the duration of the signal $2x\left(\frac{-t+c}{b}\right)$ for a=2, b=5 and c=5?

Given Answer: 60

Correct Answer: 60

QUESTION 8: MULTIPLE CHOICE

10 out of 10 points

What is energy of the signal x(t)=u(t-4)-u(t-6)-3u(t-8)+3u(t-10)?

Given Answer: 20

Correct Answer: 20

QUESTION 9: TRUE/FALSE

5 out of 5 points

rect(t).tri(t+0.5)=0 for all values of t

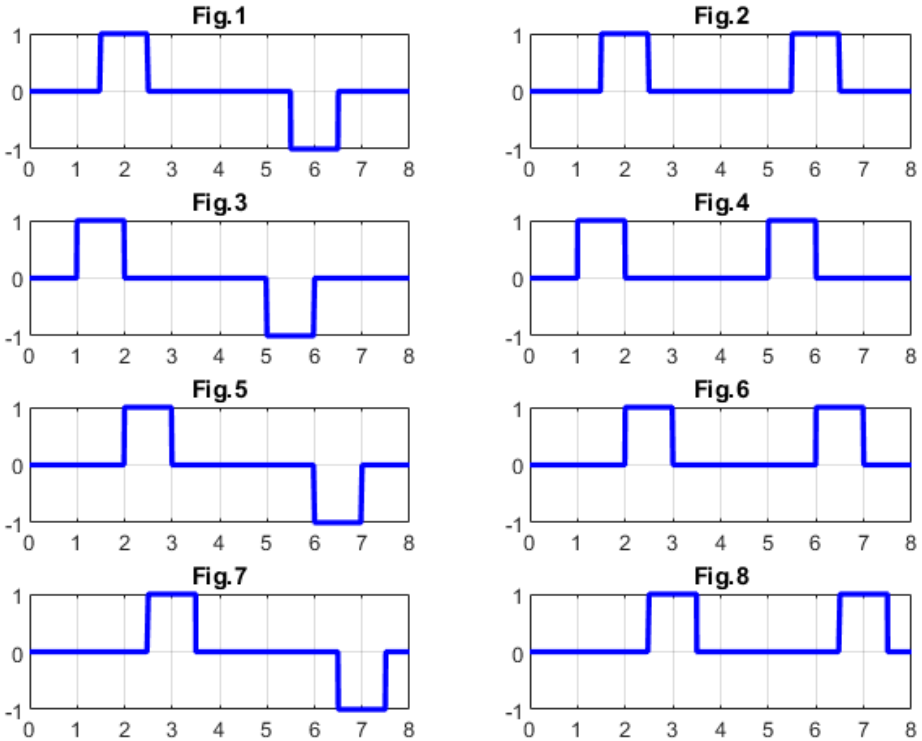
Given Answer: False

Correct Answer: False

QUESTION 10: MULTIPLE CHOICE

10 out of 10 points

A Linear and Time-Invariant system responds to the input u(t) by the output rect(t-0.5). Which one of the below figures represents the output signal when the system is excited with the signal $\mathrm{rect}\left(\frac{t-4}{4}\right)$?



Given Answer: Fig. 5

Correct Answer: Fig. 5

QUESTION 11: TRUE/FALSE

5 out of 5 points

The system $y(t) = e^{t-1}x(t+1)$ is not causal.

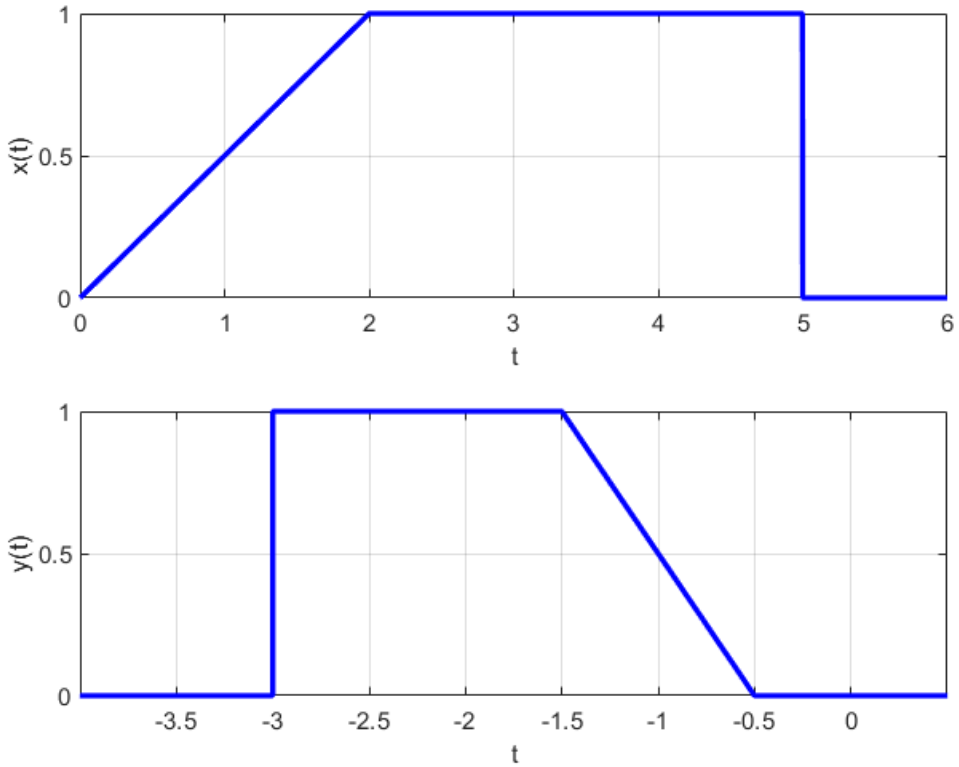
Given Answer: True

Correct Answer: True

QUESTION 12: MULTIPLE CHOICE

10 out of 10 points

Consider the signals x(t) and y(t) shown in the figure below. How are these signals related?



Given Answer: y(t)=x(-2t-1)

Correct Answer: y(t)=x(-2t-1)

QUESTION 13: CALCULATED NUMERIC

10 out of 10 points

If $\text{rect}\left(\frac{t}{10}\right) \times \text{rect}\left(\frac{t-7}{6}\right) = \text{rect}\left(\frac{t-a}{b}\right)$, what is the value of b?

Given Answer: 1

Correct Answer: 1

Answer range +/- 0 (1 - 1)